

KRISHNA Koushik Thokala

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Summary

Cloud Infrastructure Engineer with 2+ years of production NOC experience owning high-severity incidents, RCA, SLA troubleshooting, and carrier-grade reliability. Hands-on AWS/EKS, Terraform, Kubernetes, CI/CD, and observability projects focused on detecting, debugging, and recovering from cloud incidents.

Technical Skills

Cloud & AWS: VPC, IAM, EC2, S3, Lambda, API Gateway, EventBridge, CloudWatch, SNS, DynamoDB, CloudFront, EKS, ECR, RDS, ALB, Secrets Manager, Bedrock
Infrastructure & DevOps: Terraform, Docker, Kubernetes, Helm, Git, GitHub Actions, CI/CD, Linux (Ubuntu), Ingress, ConfigMaps, Secrets, Liveness/Readiness Probes, HPA
Observability & SRE: CloudWatch Logs/Metrics/Alarms, Incident Response, RCA, Runbooks, SLA Troubleshooting, MTTR, SLO Metrics, Production Operations, k6 Load Testing
Networking: TCP/IP, BGP, OSPF, MPLS, VPN, DNS, Load Balancing, Firewalls, Network Security, Juniper/Cisco Routing
Programming & Databases: Python, Bash, SQL, PostgreSQL, REST APIs, FastAPI, React
Certifications: AWS Certified Cloud Practitioner (2026)

Education

Indiana University Bloomington

Master of Science in Computer Science, GPA: 3.5/4.0

August 2024 – May 2026

Bloomington, IN

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.4/10

June 2018 – June 2022

Chennai, India

Experience

Tata Communications

Senior Engineer, Shift Lead (production NOC operations)

June 2022 – July 2024

Pune, India

- Owned high-severity production infrastructure incidents for Tier-1 telecom clients, driving triage, escalation, vendor/TAC coordination, field engineer dispatch, and service restoration across 24/7 NOC operations.
- Isolated BGP/MPLS routing failures, TCP/IP connectivity issues, DWDM optical faults, and equipment malfunctions across Juniper, Huawei, Ciena, and Alcatel platforms using alarms, logs, interface status, loop testing, and segment-wise fault isolation.
- Led 5-person global NOC shifts through 40+ daily priority incidents, including AVP/VP-level escalations, coordinating handoffs, RCA notes, customer-premises testing, and cross-functional follow-up until resolution.
- Restored high-priority services within 4-hour SLA targets by coordinating TAC teams, field engineers, vendor teams, and internal operations; earned Certificate of Excellence for carrier-grade reliability.

Projects

CloudOps SRE Platform | EKS, Terraform, Helm, RDS, GitHub Actions

June 2026

- Built an EKS-based SRE operations platform that centralizes service health, deployment history, incident timelines, MTTR, SLO tracking, and error budget views for cloud operations workflows.
- Deployed frontend/backend workloads behind ALB Ingress with Helm-managed Kubernetes deployments, services, resource limits, liveness/readiness probes, Alembic migrations, and Secrets Manager-backed database credentials.
- Added production hardening with structured JSON logs, request correlation IDs, Kubernetes manifest validation, Terraform security/cost checks, CloudWatch logs, HPA/k6 validation, and RDS secret-rotation runbooks.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

May 2026

- Architected serverless uptime monitor on AWS using EventBridge, Lambda, DynamoDB, API Gateway, SNS, CloudWatch, Terraform, and a CloudFront-hosted React dashboard.
- Optimized current-status reads from 86,400 history records to 10 latest-status rows for a 10-URL, 30-day workload; added state-change alerts, API keys, throttling, restricted CORS, KMS, DLQs, 9 CloudWatch metrics, and CI quality gates.

AWS Incident Triage Pipeline | Lambda, CloudWatch, SNS, Bedrock, Terraform

May 2026

- Engineered event-driven incident triage pipeline where CloudWatch alarm state changes trigger Lambda to parse alert context, calculate deterministic severity, and publish SNS incident notifications.
- Integrated Amazon Bedrock for remediation-focused summaries with fallback handling; provisioned EC2, Lambda, SNS, IAM, and CloudWatch alarms via Terraform with unit tests, Bandit, and Checkov CI checks.